

Plotting points and drawing graphs of relationships

Coordinates in all four quadrants, and a rule turned into a straight line.
[Cambridge insert]

LESSON 169–170

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 11 GRAPHS

LESSON CLOCK

20'

20'

25'

15'

The coordinate grid	20 min
From a rule to a table	20 min
From a table to a line	25 min
Conversion graphs	15 min

LEARNING OBJECTIVES

- Plot and read points in all four quadrants.
- Build a table of values from a rule such as $y = 2x + 1$.
- Plot the table and join the points into a straight line.
- Read values off a graph, including a conversion graph.

TEXTBOOK

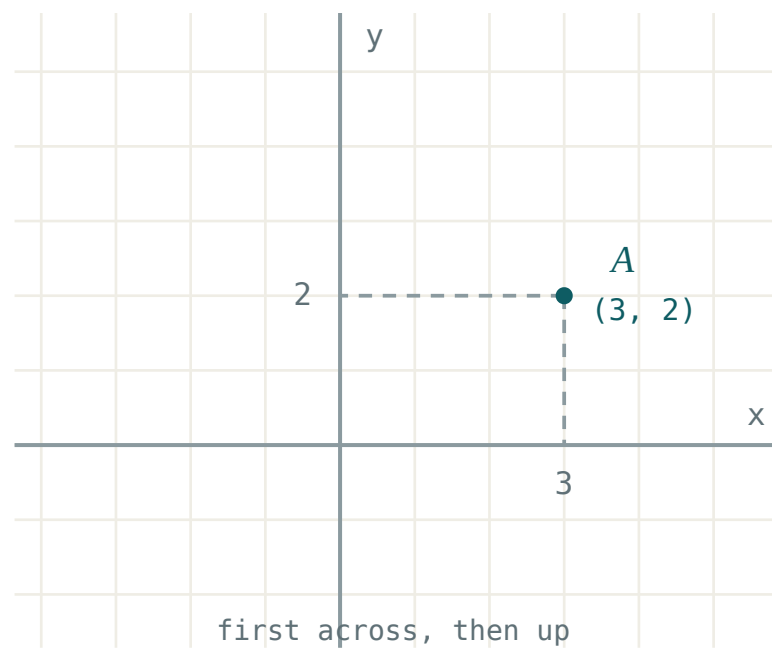
National	pp. 483–487
Cambridge	Stage 7, pp. 112–120
Workbook	Exercise 11.1–11.2

01

Explanation

The coordinate grid

Two number lines crossing at the **origin** fix every point by a pair of numbers: across first, then up. The signs decide the quadrant.



A point fixed by its two coordinates — across first, then up

POINT	SIGNS	QUADRANT	WHERE
(3, 2)	+ , +	first	right and up
(-3, 2)	- , +	second	left and up
(-3, -2)	- , -	third	left and down
(3, -2)	+ , -	fourth	right and down

(3, 2) AND (2, 3) ARE DIFFERENT POINTS

The order is fixed: the first number is the x -coordinate, measured along, and the second is the y -coordinate, measured up. “Along the corridor, then up the stairs.”

From a rule to a table

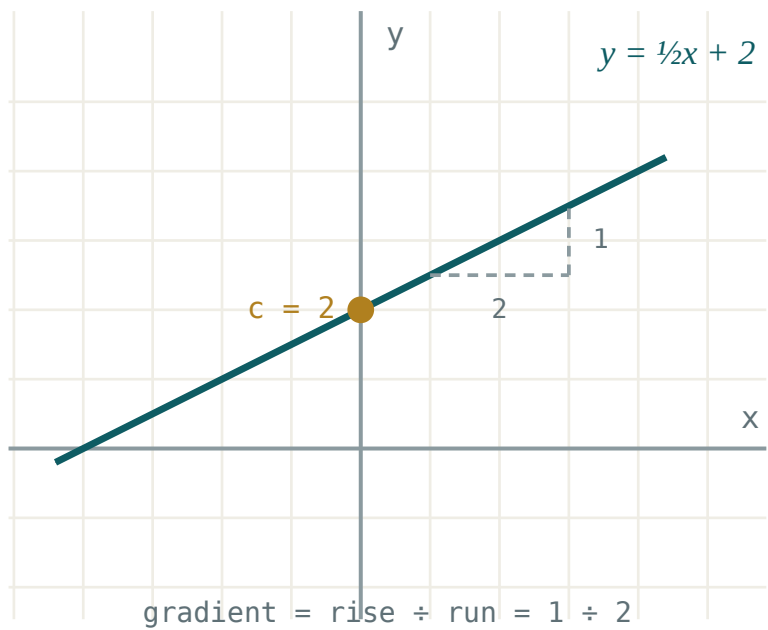
A rule connects two quantities. Choose values of x , work out each y , and collect the pairs in a table.

X	-2	-1	0	1	2	3
$y = 2x + 1$	-3	-1	1	3	5	7
$y = x + 3$	1	2	3	4	5	6
$y = 10 - x$	12	11	10	9	8	7

EQUAL STEPS IN x GIVE EQUAL STEPS IN y

In the first row every step of 1 across raises y by 2 ; in the last it lowers y by 1 . That steadiness is exactly why the points will lie on a straight line.

From a table to a line



A straight-line graph — the steady climb and the point where it crosses the y -axis

STEP	WHAT TO DO	WHY
1	draw and number both axes, including the negatives	every point must fit
2	plot each pair from the table with a small cross	a dot hides its own centre
3	check the points look like a line before joining	one stray point is an arithmetic slip
4	join them with a ruler and extend a little each way	the rule holds beyond the table
5	label the line with its rule	two lines on one grid must be told apart

A POINT OUT OF LINE IS A MISTAKE, NOT A FEATURE

Recalculate that y rather than drawing a bend. The rules in this topic all give perfectly straight lines.

Conversion graphs

A straight-line graph is the fastest way to convert between two units. Eight kilometres is about five miles, so the line through $(0, 0)$ and $(8, 5)$ converts either way.

KILOMETRES	8	16	24	40
miles	5	10	15	25

Not every real-life graph starts at the origin. A taxi charging 5000 so'm plus 2000 so'm a kilometre has the rule $c = 5000 + 2000d$.

KILOMETRES d	0	1	3	5	7
cost in so'm	5000	7000	11 000	15 000	19 000

READ ACROSS, THEN DOWN – OR DOWN, THEN ACROSS

To convert 24 km, go up from 24 to the line and across to 15 miles. To go the other way, start on the miles axis. One line answers both questions.

02 Worked examples

EXAMPLE 1 Complete a table for $y = 2x + 1$ from $x = -2$ to $x = 3$.

- ① Double each x and add 1.

- ② $-3, -1, 1, 3, 5, 7$.

- ③ Each step of 1 in x raises y by 2.
So the points lie on a line ✓

ANSWER

$-3, -1, 1, 3, 5, 7$

EXAMPLE 2 Using $8 \text{ km} = 5 \text{ miles}$, convert 24 km and 20 miles.

- ① 24 km is three lots of 8.

- ② So $3 \cdot 5 = 15$ miles.

- ③ 20 miles is four lots of 5, so $4 \cdot 8 = 32$ km.
The same line, read both ways.

ANSWER

15 miles and 32 km

EXAMPLE 3 A taxi charges 5000 so‘m plus 2000 so‘m a kilometre. A ride costs 19 000 so‘m. How far was it?

① $19\,000 - 5000 = 14\,000$ so‘m for the distance.

② $14\,000 \div 2000$.

③ $= 7$ km.

Check: $5000 + 14\,000 = 19\,000$ ✓

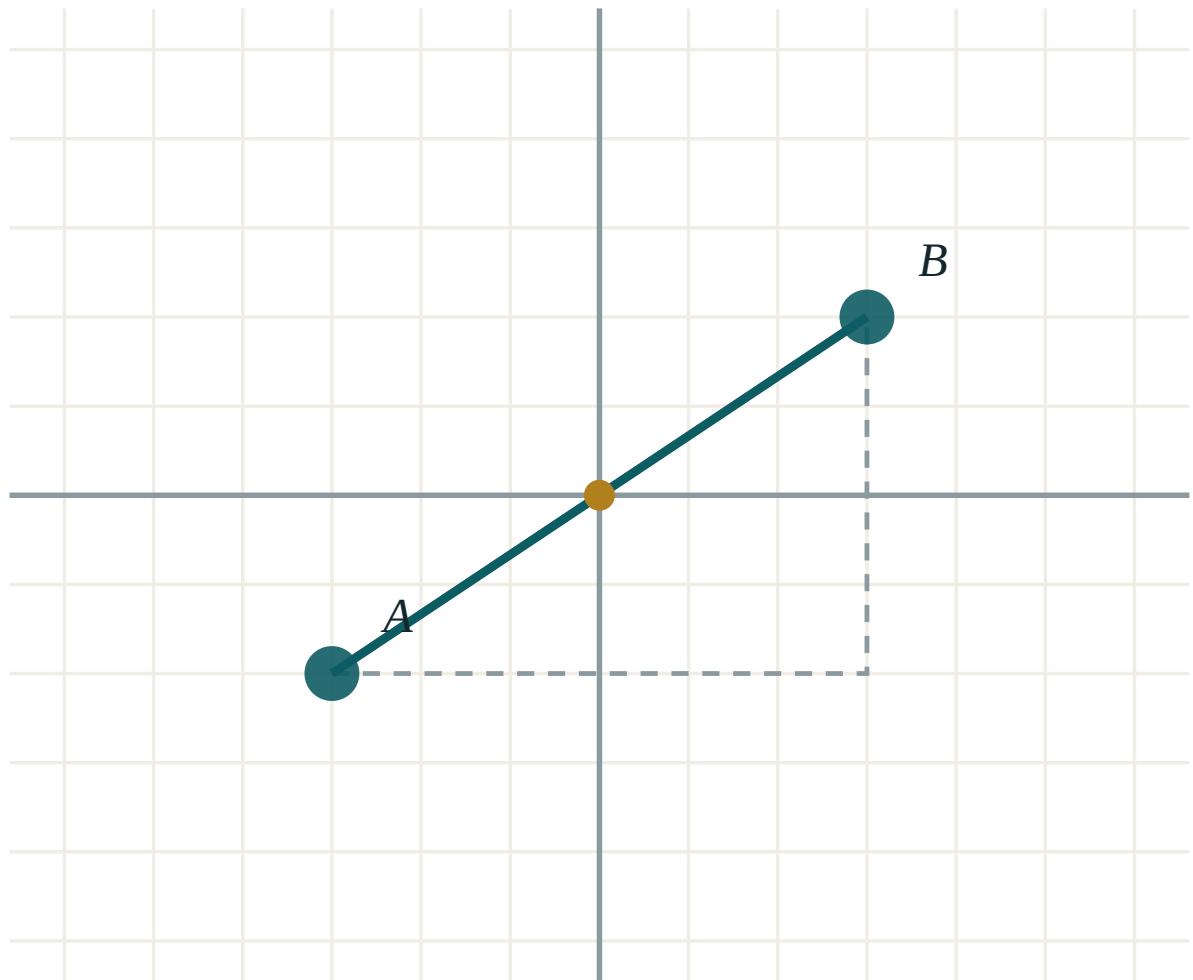
ANSWER

7 km

03 Interactive model

Drag one point in the model along a row and watch only the first coordinate change; that is the whole of “across first, then up”. The distance readout is a Grade 8 preview — ignore it for now.

Points on the coordinate grid



A (-3, -2)

B (3, 2)

 $x_2 - x_1$ 6 $y_2 - y_1$ 4

distance AB 7.21

midpoint M (0, 0)

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Coordinates	Koordinatalar	Координаты
Origin	Koordinata boshi	Начало координат
x-axis	Abssissa o'qi	Ось абсцисс
y-axis	Ordinata o'qi	Ось ординат
Quadrant	Chorak	Четверть
Table of values	Qiyimatlar jadvali	Таблица значений
Straight-line graph	To'g'ri chiziqli grafik	Прямолинейный график
Conversion graph	O'tkazish grafigi	График перевода

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

In $(3, -2)$ the -2 is measured:

along

up or down

diagonally

from B

$(-3, 2)$ lies in the:

first quadrant

second quadrant

third quadrant

fourth quadrant

$y = 2x + 1$ at $x = 3$ is:

5

7

6

9

A graph crosses the y -axis where:

Points from $y = 2x + 1$ lie on:

With $8 \text{ km} = 5 \text{ miles}$, 40 km is:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. The quadrant of $(3, 2)$

ANSWER The first

2. The quadrant of $(-3, 2)$

ANSWER The second

3. The quadrant of $(-3, -2)$

ANSWER The third

4. The quadrant of $(3, -2)$

ANSWER The fourth

5. $y = 2x + 1$ at $x = 3$

ANSWER 7

6. $y = 2x + 1$ at $x = 0$

ANSWER 1

7. On the y -axis the x -coordinate is

ANSWER 0

Medium · the standard question

7 PROBLEMS

1. $y = 2x + 1$ for $x = -2, -1, 0, 1$

ANSWER $-3, -1, 1, 3$

2. $y = x + 3$ at $x = -2$

ANSWER 1

3. $y = 3x - 2$ at $x = 2$

ANSWER 4

4. $y = 10 - x$ at $x = 4$

ANSWER 6

5. With $8 \text{ km} = 5 \text{ miles}$, 16 km

ANSWER 10 miles

6. ...and 40 km

ANSWER 25 miles

7. A taxi at $5000 + 2000d$ for 3 km

ANSWER $11\,000 \text{ so'm}$

Hard · stretch and reason

7 PROBLEMS

1. $y = 2x + 1$: the value of x when $y = 11$

ANSWER $x = 5$

2. $y = 3x - 2$: the value of x when $y = 13$

ANSWER $x = 5$

3. With $8 \text{ km} = 5 \text{ miles}$, 24 km

ANSWER 15 miles

4. ...and 20 miles

ANSWER 32 km

5. A taxi ride at $5000 + 2000d$ costs $19\,000$ so'm: the distance

ANSWER 7 km

6. Why do the points of $y = 2x + 1$ lie on a line?

ANSWER Equal steps in x give equal steps in y

7. The graph of $y = 10 - x$ slopes

ANSWER Downwards

07 Homework

Homework — 5 tasks

Plot with small crosses, join with a ruler, and label each line with its rule.

- Plot $(2, 3)$, $(-2, 3)$, $(-2, -3)$ and $(2, -3)$, and name the shape they make.
- Complete a table for $y = 3x - 1$ from $x = -2$ to $x = 3$, and draw the graph.
- On the same grid draw $y = x + 3$ and write down where the two lines cross.
- Using $8 \text{ km} = 5 \text{ miles}$, convert 32 km and 15 miles .

5. A taxi charges 4000 so'm plus 1500 so'm a kilometre. Find the cost of 6 km and the distance costing 16 000 so'm.